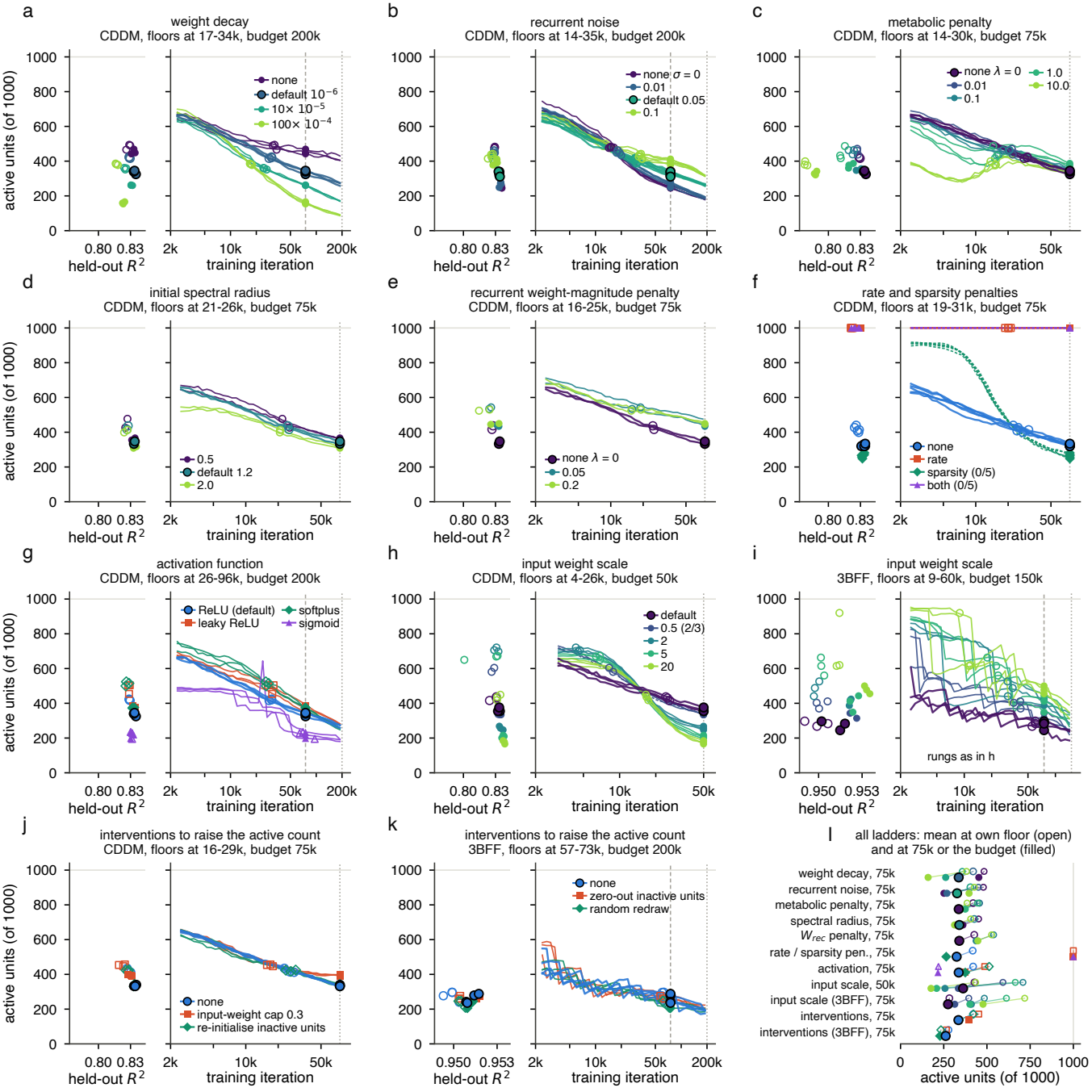

a
weight decay
CDDM, floors at 17-34k, budget 200k
none
default 10^{-6}
$10\times\ 10^{-5}$
$100\times\ 10^{-4}$
b
recurrent noise
CDDM, floors at 14-35k, budget 200k
none $\sigma = 0$
0.01
default 0.05
0.1
c
metabolic penalty
CDDM, floors at 14-30k, budget 75k
none $\lambda = 0$
0.01
0.1
1.0
10.0
d
initial spectral radius
CDDM, floors at 21-26k, budget 75k
0.5
default 1.2
2.0
e
recurrent weight-magnitude penalty
CDDM, floors at 16-25k, budget 75k
none $\lambda = 0$
0.05
0.2
f
rate and sparsity penalties
CDDM, floors at 19-31k, budget 75k
none
rate
sparsity (0/5)
both (0/5)
g
activation function
CDDM, floors at 26-96k, budget 200k
ReLU (default)
leaky ReLU
softplus
sigmoid
h
input weight scale
CDDM, floors at 4-26k, budget 50k
default
0.5 (2/3)
2
5
20
i
input weight scale
3BFF, floors at 9-60k, budget 150k
rungs as in h
j
interventions to raise the active count
CDDM, floors at 16-29k, budget 75k
none
input-weight cap 0.3
re-initialise inactive units
k
interventions to raise the active count
3BFF, floors at 57-73k, budget 200k
none
zero-out inactive units
random redraw
l
all ladders: mean at own floor (open)
and at 75k or the budget (filled)
weight decay, 75k
recurrent noise, 75k
metabolic penalty, 75k
spectral radius, 75k
W_{rec} penalty, 75k
rate / sparsity pen., 75k
activation, 75k
input scale, 50k
input scale (3BFF), 75k
interventions, 75k
interventions (3BFF), 75k
active units (of 1000)
held-out R^2
training iteration